

SciOdyssey.

A research layer over your agent harness.

PRESERVE THE RESEARCH WORLD · RESET THE RESEARCHER

ROBUST

SUPERVISABLE

BROAD SEARCH

SAME EVIDENCE
FRESH JUDGMENT
A WIDER SEARCH

Research
Runs
Longer.

01 / PROBLEM

Why current agents fall short.

Sustained autonomous exploration encounters three critical bottlenecks:

01 Closed-world fragility

Rigid predefined pipelines shatter as soon as unexpected runtime errors emerge. In Cycle 1, a **float32 JSON serialization bug** halted logging; **without autonomous self-healing, exploration dies permanently**.

02 Single-trajectory inertia

Accumulating context history anchors the agent to past reasoning, **trapping search in local basins** (e.g., standard single-agent baselines **stall at 0.6044** while endlessly fine-tuning hyperparameters).

03 Context inflation cost

Continually rereading deep conversational context **costs 8x more tokens** than the actual code edits, exhausting context windows and causing **severe reasoning dilution**.

TRADITIONAL AGENT HARNESS

Single Monolithic Context · Error Cascades · Token Waste · Local Optima

SCIODYSSEY RESEARCH LAYER

Ephemeral Fresh Scientist · Persistent Disk Memory · Verified Handoff

PARADIGM SHIFT

Decouple the **persistent research world** (disk memory) from ephemeral **researcher reasoning** (fresh context).

02 / SYSTEM

One persistent world. Fresh scientific judgment.

Give the research world and the researcher different lifetimes.

THE DUAL-LIFETIME ENGINE

Context reset between trajectories · Audited persistence on disk

01 Environment

Task contract, data inputs, official evaluator & compute budget. Read-only ground truth.

02 Research World

Verified ledger, working priors, and reproducible code. Git worktree per branch.

03 Fresh Scientist

GPT ↔ Gemini rotation: clean hypotheses, code & experiments. Ephemeral reasoning.

04 META Review

Audit claims ↔ telemetry evidence; crystallize verified State. Strict promotion gate.

4-CYCLE MODEL RELAY

C1 GPT-5.6 (Baseline)

C2 Gemini-3.7 (Features)

C3 GPT-5.6 (Variance)

C4 Gemini-3.7 (Ensemble)

Gemini META throughout

Trajectory Handoff Loop: Fresh Scientist reset per cycle; only **audited empirical findings** enter durable disk memory.

PERSISTENT RESEARCH WORLD

What survives across trajectories

A · VERIFIED EVIDENCE

Knowledge + Ledger

Complete metric trajectories, traceback failure logs, and full report artifacts stored in **ledger.jsonl**.

B · WORKING PRIORS

Brief + Hypotheses

Current conceptual understanding, domain beliefs, and exploration leads in **brief.md**, open to active revision.

C · REPRODUCIBLE STATE

Retained Code + State

Tested pipeline versions, serialized model weights, starter kit checkers, and full **git commit provenance**.

WORKTREE ISOLATION

Physical Git worktrees isolate state per branch; prevents concurrent file collisions; automated CI/CD artifact capture.

AUTONOMOUS SELF-HEALING

Runtime traceback interception → Automated patch generation → Targeted regression testing → Safe state rollback.

MULTI-MODEL SYNTHESIS

Reviewer synthesizes cross-branch findings; merges complementary features & model architectures post-hoc.

The next Scientist inherits **verified experience, not prior reasoning inertia**.

Decoupled Memory · Audited Persistence · Autonomous Self-Healing · Zero Context Pollution

03 / EVALUATION

Five empirical claims. Measured evidence.

Standardized evaluation on KuaiRand-Pure benchmark · Fixed official evaluator · Complete telemetry accounting

01 MEASURED RESULT

The model got better.

0.6016000

Official FM baseline

→

0.6059363

SciOdyssey submission

+0.0043363 Primary improvement

GAUC 0.6728421

(+0.0054)

nDCG@5 0.5390304

(+0.0033)

LogLoss 0.38102

(-0.0032)

Feature Engine: 46 categorical cross-interactions + log1p transforms

Architecture: 8-seed Factorization Machine ensemble (NumPy / CPU)

Monotonic gain from baseline (0.6016) → feature engineering (0.6041) → seed ensembling (0.6059); **100% CPU inference** without GPU clusters.

02 SUSTAINED SEARCH

The search stayed productive.

0.60163

C1: Valid Base

0.60409

C2: Features

0.60443

C3: Var. Red.

0.60594

C4: 8-Seed FM

SciOdyssey	0.6059363	Climbed across 4 cycles (7 Full evals)
Codex Goal	0.6044533	Peaked at C8; stalled & dropped in 22 runs

4 autonomous cycles

7 Full evaluations

+0.00148 over Codex

7 Full public-validation evaluations retained across 4 cycles. Trajectory climbed via **structural feature expansion** rather than single lucky edits.

04 RESOURCE ACCOUNTING

We are not tokenmaxxing.

PROJECT COST

~\$10

First code to final result

HARDWARE

1 × M2

1 consumer laptop

COMPUTE

0 GPU-hr

100% CPU / NumPy only

AGENT RUNTIME

1h 55m

Wall-clock across 4 cycles

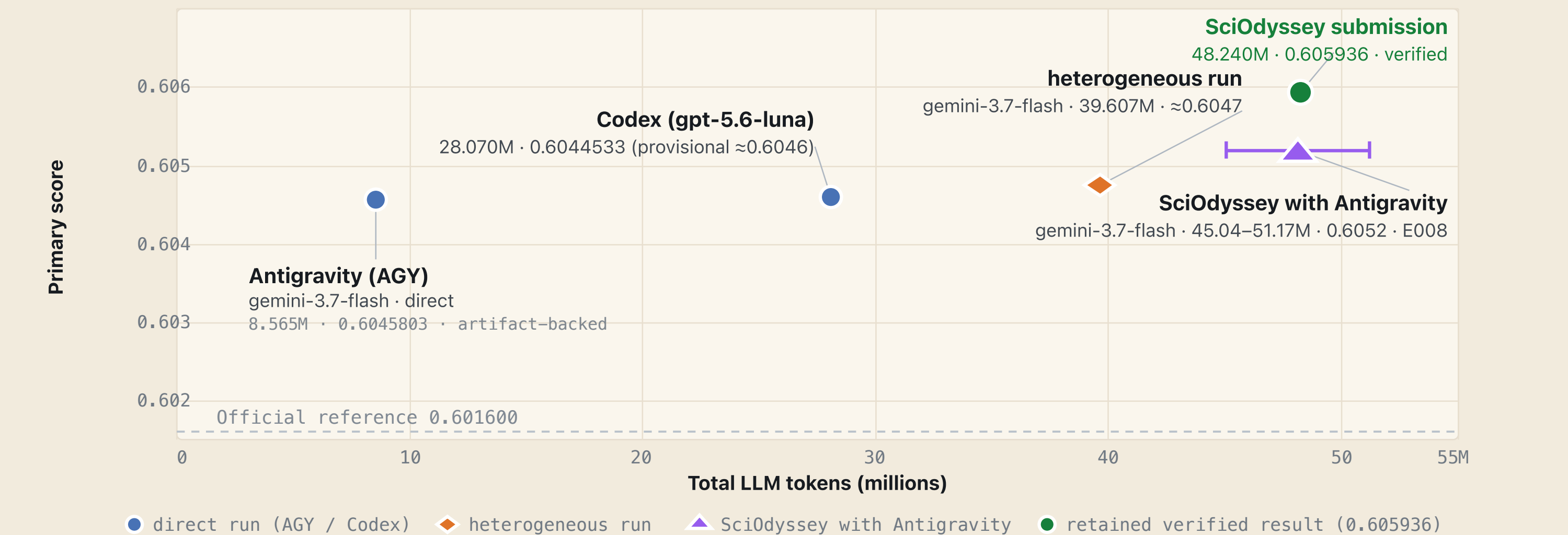
Performance came from the autonomous research loop, not an expensive compute budget. Full discovery executed on consumer hardware without GPU clusters.

05 DIRECT AGENT BENCHMARK

Score versus measured LLM-token investment.

KuaiRand-Pure Public Validation

Input + output including cache-read · Whole recorded agent system



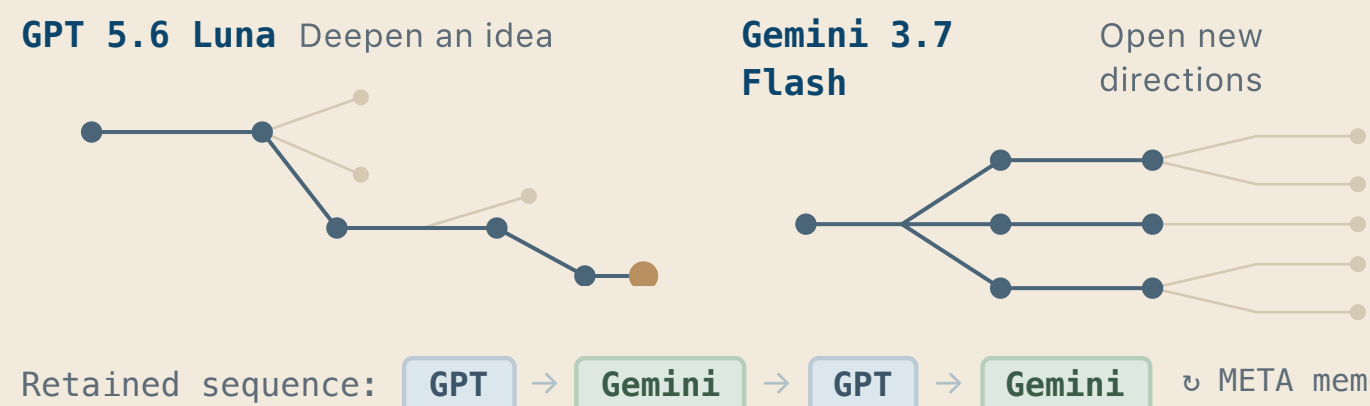
All results **verified** on KuaiRand-Pure official validation split with **complete reproducible artifacts**.

04 / INSIGHTS

What the run taught us.

01 · MODEL DIVERSITY

Depth + Breadth Synergy.



Gemini-only 0.6052 → GPT+Gemini 0.6059363

02 · PARALLEL SYNTHESIS

Cross-Branch Idea Transfer.



Lower-scoring Branch C contributed pairwise ranking and rank normalisation — a lower score can still carry a useful idea.

Branch B+C Ideas → Synthesis 0.605536

05 / EXPERIENCE

Start your journey.

task.md

What to solve

Formal task contract: research objectives, dataset inputs, compute constraints, and official evaluation protocol.

- Dataset paths & split integrity
- Official sandbox evaluator

PERSONAL.md

How to work

User research preferences, model guidance, exploration style, tool permissions, and supervisory checkpoints.

- Model rotation (GPT / Gemini)
- Supervision & budget thresholds

SUPERVISION TIERS

Tier 1 Unattended Overnight Runs · Tier 2 Checkpoint Review (META) · Tier 3 Steer Working Priors

1. Define task.md → 2. Single-step test → 3. Autonomous search → 4. Parallel explore → 5. Inspect evidence

DECLARATIVE PROTOCOL

Human defines boundary and preferences (task.md + PERSONAL.md); SciOdyssey autonomously iterates, heals, and crystallizes verified evidence.

github.com/zc6600/research-agent-kuairand

Team Good4AI: Chen Zhu · Zhou Ziyu · Shilin Xu · GE GAO · Jiran Li

Sources: slides/slides.md · docs/ARCHITECTURE.md · docs/FINAL_REPORT.md · KuaiRand-Pure submission ledger

TEAM GOOD4AI · ROBUST · SUPERVISABLE · BROAD SEARCH

A Persistent Research World for Autonomous Science

PROJECT WIKI & DEMO

sciodyyssey.zc6600.wiki

Scan for live report & artifacts

